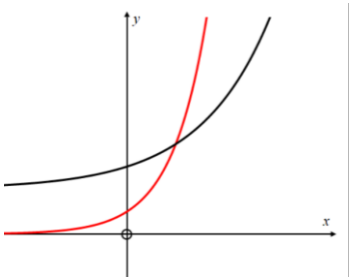
Question			Answer	Marks	AO	Guidance
9	(a)		Stretch in the x-direction Stretch scale factor ½	B1 B1	1.2 1.1b	
				[2]		
9	(b)			B1 B1	1.1b 1.1b	General shape of exponential graph less steep than the given graph for positive x (note red graph is printed) Horizontal asymptote above the x-axis and intersection with y-axis must be above that for the given graph
				[2]		
9	(c)		The graphs intersect when $e^{2x} = k + e^x$ So when $e^{2x} - e^x - k = 0$ discriminant $(-1)^2 - 4(-k)$ is negative for $k < -\frac{1}{4}$ so no real roots and no points of intersection	M1 M1 E1	2.1 2.1 2.1	Attempts to solve simultaneously. Allow $k = -\frac{1}{4}$ substituted Uses discriminant of the equation must state no real roots or no points of intersection
				[3]		
9	(d)		When $k = 2$, $e^{2x} - e^x - 2 = 0$ gives $e^x = -1, 2$ So $x = \ln 2$ as $e^x = -1$ is not possible	M1 A1	2.1 2.1	Evaluates e^x from their quadratic and attempts to use natural logs must state that ln2 is a root and that there are no others. Allow SC1 for substituting $x = \ln 2$ in both equations and concluding it must be a root

Question			Answer	Marks	AO	Guidance
				[2]		

Question	Answer	Marks	AO	Guidance